

Use Case Scenario:

The following table summarizes the use cases of the system.

Level – 0	Level – 1	Level – 2	Actors
Circulation System	Authentication	Sign Up	Student, Faculty
		Sign In	Administrator, Student, Faculty, Librarian
		Sign Out	Administrator, Student, Faculty, Librarian
		Change Passwords	Administrator, Student, Faculty
	Configure	Change User Types	Administrator, Librarian
		Configure the Due Date for an Item	Administrator
		Configure the Fine for Overdue Items	Administrator
		Change default Item Due dates	Librarian
	Update	Add an Item	Librarian
		Edit an Item	Librarian
		Delete an Item	Librarian
	Return	Retrieve an Item	Librarian
		Reports on Over Due Items	Student, Faculty
	Borrow	Search for Items	Librarian, Student, Faculty, Administrator
		Issue an Item	Librarian
		Renew Items	Student, Faculty
		Booking Items	Students, Faculty

Use Case Descriptions:

01.Authentication

i.

Use case: Sign Up

Primary Actors: Student, Faculty

Goal in context: To register in the system

Precondition:

1. System has been programmed for add new user in database
2. System has interface for registration

Triggers: The student and faculty has a need to register

Scenario:

1. Visit the register page
2. Input required information
3. Check availability for username & check validity of Password
4. Authentication and Robot checking
5. E-mail sent to user e-mail address
6. User confirm from his/ her e-mail address
7. Confirmation message showed

Exception:

1. User in not authorized for registration
2. Ambiguous Input
3. Authentication Fail

Priority: Essential, must be implemented

When Available: First increment

ii.

Use case: Sign In

Primary Actors: Student, Faculty, Administrator, Librarian

Goal in context: To enter the system

Precondition: Must be registered

Triggers: Need to log in the system

Scenario:

1. Visit the login page
2. Input Username & Password
3. Proceed to the next activity

Exception:

1. Unrecognized Username
2. Wrong Password
3. User is blocked

Priority: Essential, must be implemented

When Available: First increment

iii.

Use case: Sign Out

Primary Actors: Student, Faculty, Administrator, Librarian

Goal in context: To exit from the system

Precondition: Must be logged in

Triggers: Need to log out from the system

Scenario:

1. Click the logout button

Priority: Essential, must be implemented

When Available: First increment

iv.

Use case: Change Password(s)

Primary Actors: Student, Faculty

Goal in context: To change the existing password to a new password

Precondition: System has been programmed for a password

Triggers: The student and faculty has a need to change the existing password to a new one

Scenario:

2. Visit the login page and login
3. Click on Edit button
4. Change Password
5. Proceed to the next activity

Exception: Weak Password: Password length is too short

Priority: Essential, must be implemented

When Available: First increment

02.Configure:

i.

Use case: Change User Type(s)

Primary Actors: Administrator, Librarian

Goal in context: To change the user type

Precondition: Must be logged in as Administrator/ librarian

Triggers: The administrator and librarian have a need to change the user type.

Scenario:

1. Visit Login page and Log in
2. Click the Edit User button
3. Select the User
4. Click on the Edit button
5. Change the type for the selected User
6. Proceed to the next activity

Exception:

1. Invalid User: User may not be eligible for that type
2. Unrecognized: User does not exist

Priority: Essential, must be implemented

When Available: First increment

ii.

Use case: Configure the Due Date for an Item(s)

Primary Actors: Administrator

Goal in context: To configure the due date for an item

Precondition:

1. Must be logged in as Administrator
2. System has been programmed for editing due date

Triggers: The administrator has a need to configure the due date for an item.

Scenario:

1. Visit Login page and Log in
2. Click on Maintain Item button
3. Select the Item
4. Click on Edit Due Date for an Item button
5. Change the Due Date for selected Item
6. Proceed to the next activity

Exception:

1. Item Unavailable: Requested item does not exist
2. Ambiguous Input

Priority: Expected

When Available: Second increment

iii.

Use case: Configure the Fine for Overdue Item(s)

Primary Actors: Administrator

Goal in context: To configure the fine for overdue item

Precondition:

1. Must be logged in as Administrator
2. System has been programmed for editing due fine

Triggers: The administrator has a need to configure the fine for overdue item.

Scenario:

1. Visit Login page and Log in
2. Click on Maintain Validation Data button
3. Click on Configure the Fine for an Overdue Item button
4. Update the Fine for selected Overdue Item
5. Click on Update button
6. Proceed to the next activity

Exception: Ambiguous Input

Priority: Expected

When Available: Second increment

iv.

Use case: Change default Due date for Item

Primary Actors: Librarian

Goal in context: To change the default due date for item

Precondition: Must be logged in as Librarian

Triggers: The librarian has a need to change the default due date for item.

Scenario:

1. Visit Login page and Log in
2. Click on Maintain Item button
3. Click on Change default Due date for Item button
4. Update the Due date for selected Item
5. Proceed to the next activity

Exception: Ambiguous Input

Priority: Expected

When Available: Second increment

03.Update

i.

Use case: Add an Item(s)

Primary Actors: Librarian

Goal in context: To add new item(s)

Precondition:

1. System has been programmed for adding item in database
2. Must be logged in as Librarian

Trigger: The librarian has a need to add new item(s)

Scenario:

1. Visit Login page and Log in
2. Click on Maintain Item button
3. Click on Add Item button to add new item
4. Enter the new Item data (select Location) and confirm changes
5. Proceed to the next activity

Exception: Already Exist: Requested item is already added in the database

Priority: Essential, must be implemented

When Available: First increment

ii.

Use case: Edit an Item(s)

Primary Actors: Librarian

Goal in context: To edit an item

Precondition:

1. System has been programmed for editing item in database
2. Must be logged in as Librarian

Trigger: The librarian has a need to edit an item(s).

Scenario:

1. Visit Login page and Log in
2. Click on Maintain Item button
3. Search and Select the Item to edit
4. Click on Edit Item button
5. Edit the Item details and confirm changes
6. Proceed to the next activity

Exception:

1. Does not exist: Requested item does not exist in the database
2. Ambiguous Input

Priority: Essential, must be implemented

When Available: First increment

iii.

Use case: Delete an item(s)

Primary Actors: Librarian

Goal in context: To delete an item

Precondition:

1. System has been programmed for deleting item in database
2. Must be logged in as Librarian

Trigger: The librarian has a need to delete an item(s).

Scenario:

1. Visit Login page and Log in
2. Click on Maintain Item button
3. Search and Select the Item to delete
4. Click on Delete Item button
5. Delete the selected Item and confirm changes
6. Proceed to the next activity

Exception: Does not exist: Requested item does not exist in the database

Priority: Essential, must be implemented

When Available: First increment

04.Return

i.

Use case: Retrieve an Item(s)

Primary Actors: Librarian

Goal in context: To retrieve an item

Precondition: Item must be Issued for the particular user

Trigger: The librarian has a need to retrieve an item

Scenario:

1. Visit Login page and Log in
2. Click on Retrieve button
3. Enter User name and Click Search for user
4. Select the User
5. Enter Item Name and Click Search item
6. Select the Item
7. Click Retriever button and Status changes from Issued to Retrieved
8. A message is displayed
9. Proceed to the next activity

Priority: Essential, must be implemented

When Available: First increment

ii.

Use Case: Reports on Over Due Item(s)

Primary Actors: Student, Faculty

Goal in context: To generate reports on overdue item(s)

Precondition:

1. System has been programmed for automated report generation
2. Must be logged in as Librarian

Triggers: Report need to be automatically generated of overdue item(s).

Scenario:

1. Database is automatically checked daily for Over Due
2. Send mail to users having Over Due
3. Increase fine if not cleared
4. Update database
5. Proceed to the next activity

Exception: Error: System is not ready

Priority: Essential, must be implemented

When Available: First increment

05.Borrow

i.

Use case: Search for Item(s)

Primary Actors: Librarian, Student, Faculty, Administrator

Goal in context: To perform a search for item(s)

Precondition: System has been programmed for searching all items in database

Triggers: The student, faculty, librarian, administrator has a need to search for item(s)

Scenario:

1. Visit the main page
2. Enter data and information such as title, author's name etc.
3. Click the Search button
4. View the search result
5. Proceed to the next activity

Exception:

1. Search item does not exist
2. User is not eligible for searching that item

Priority: Essential, must be implemented

When Available: First increment

ii.

Use case: Issue an Item(s)

Primary Actors: Librarian

Goal in context: To issue an item

Precondition:

1. User must be eligible for taking requested item

2. Item is available

Trigger: The librarian has a need to issue an item.

Scenario:

1. Visit Login page and Log in
2. Click on Issue button
3. Search for the Person/User
4. Select User and list of items issued to the user is displayed
5. Click on Issue button and Status changes from Available to Issued to Library
6. A message is displayed
7. Proceed to the next activity

Exception:

1. Invalid User: User status is not supported for this event
2. Item does not exist

Priority: Essential, must be implemented

When Available: First increment

iii.

Use case: Renew Item(s)

Primary Actors: Student, Faculty

Goal in context: To extend the item(s) that has/have reached the due date.

Precondition:

1. Valid User
2. Valid and Available Item

Triggers: The student, faculty, librarian has a need to extend item(s) due date.

Scenario:

1. Visit the login page and log in
2. List of Issued items for that User will be displayed
3. Extend due date button will be shown for all the items which were not renewed in the past

4. Click on the Extend due date button to renew the item
5. Item(s) due date will be extended for 14 days

Exception:

1. Time Limit Exceeded: Renew chance has been finished
2. Unavailable: Item is unavailable for renew

Priority: Essential, must be implemented

When Available: First increment

iv.

Use case: Booking Item(s)

Primary Actors: Student, Faculty

Goal in context: To book item(s) that are unavailable at that particular time.

Precondition:

1. Valid User
2. Valid but unavailable Item at the particular time

Triggers: The student and faculty have a need to book item(s) that is/are unavailable at that particular time.

Scenario:

1. Visit the login page and log in
2. Enter data and information such as title, author's name etc. or
3. Click the Search button
4. View the search result
5. Click on the Booking button
6. Proceed to the next activity

Exception: Missing: Item is missing

Priority: Essential, must be implemented

When Available: First increment